

ECEN 629 Convex Optimization for Electrical Engineering

Chapter 10 Equality constrained minimization

Instructor: Dr. Chao TIAN

Texas A&M University-College Station

Equality constrained minimization

$$\begin{array}{ll}\text{minimize} & f(x) \\ \text{subject to} & Ax = b\end{array}$$

- f convex, twice continuously differentiable
- $A \in \mathbf{R}^{p \times n}$ with $\text{rank } A = p$
- we assume p^* is finite and attained

optimality conditions: x^* is optimal iff there exists a ν^* such that

$$\nabla f(x^*) + A^T \nu^* = 0, \quad Ax^* = b$$

How to solve equality constrained minimization

For the KKT equations:

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Finding optimal solutions = Solving KKT condition equations.

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If it is difficult to directly solve these equations:

- Convert an equality constrained problem to an unconstrained problem;
- Solve the dual problem (if the dual is simple and twice differentiable)
- Newton's method to directly work on the equality constrained problem:
main practical advantage is to retain the original structure of the problem.

equality constrained quadratic minimization (with $P \in \mathbf{S}_+^n$)

$$\begin{array}{ll} \text{minimize} & (1/2)x^T P x + q^T x + r \\ \text{subject to} & Ax = b \end{array}$$

optimality condition:

$$\begin{bmatrix} P & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} x^* \\ \nu^* \end{bmatrix} = \begin{bmatrix} -q \\ b \end{bmatrix}$$

- coefficient matrix is called KKT matrix
- KKT matrix is nonsingular if and only if

$$Ax = 0, \quad x \neq 0 \quad \implies \quad x^T P x > 0$$

- equivalent condition for nonsingularity: $P + A^T A \succ 0$

Eliminating equality constraints

represent solution of $\{x \mid Ax = b\}$ as

$$\{x \mid Ax = b\} = \{Fz + \hat{x} \mid z \in \mathbf{R}^{n-p}\}$$

- $\hat{x}$ is (any) particular solution
- range of $F \in \mathbf{R}^{n \times (n-p)}$ is nullspace of A ($\text{rank } F = n - p$ and $AF = 0$)

reduced or eliminated problem

$$\text{minimize } f(Fz + \hat{x})$$

- an unconstrained problem with variable $z \in \mathbf{R}^{n-p}$
- from solution z^* , obtain x^* and ν^* as

$$x^* = Fz^* + \hat{x}, \quad \nu^* = -(AA^T)^{-1}A\nabla f(x^*)$$

example: optimal allocation with resource constraint

$$\begin{array}{ll}\text{minimize} & f_1(x_1) + f_2(x_2) + \cdots + f_n(x_n) \\ \text{subject to} & x_1 + x_2 + \cdots + x_n = b\end{array}$$

eliminate $x_n = b - x_1 - \cdots - x_{n-1}$, *i.e.*, choose

$$\hat{x} = be_n, \quad F = \begin{bmatrix} I \\ -\mathbf{1}^T \end{bmatrix} \in \mathbf{R}^{n \times (n-1)}$$

reduced problem:

$$\text{minimize} \quad f_1(x_1) + \cdots + f_{n-1}(x_{n-1}) + f_n(b - x_1 - \cdots - x_{n-1})$$

(variables $x_1, \dots, x_{n-1}$)

Solving equality constrained problems via the dual

Recall the dual

$$g(v) = -b^T v - f^*(-A^T v).$$

where $f^*(\cdot)$ is the conjugate of f .

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The dual problem is

$$\text{maximization: } -b^T v - f^*(-A^T v),$$

and here strong duality holds $\Rightarrow$ solving the dual gives the primal solution.

Example: Equality constrained analytic center

Consider the problem:

$$\text{minimize: } f(x) = - \sum_{i=1}^n \log x_i$$

$$\text{subject to: } Ax = b,$$

where $A \in \mathbb{R}^{p \times n}$, and $x \succ 0$. The dual is

$$\text{maximize: } g(v) = -b^T v + n + \sum_{i=1}^n \log(A^T v)_i,$$

where $A^T v \succ 0$ (implicit constraint).

Newton step

Newton step Δx_{nt} of f at feasible x is given by solution v of

$$\begin{bmatrix} \nabla^2 f(x) & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} -\nabla f(x) \\ 0 \end{bmatrix}$$

interpretations

- Δx_{nt} solves second order approximation (with variable v)

$$\begin{array}{ll} \text{minimize} & \hat{f}(x+v) = f(x) + \nabla f(x)^T v + (1/2)v^T \nabla^2 f(x)v \\ \text{subject to} & A(x+v) = b \end{array}$$

- Δx_{nt} equations follow from linearizing optimality conditions

$$\nabla f(x+v) + A^T w \approx \nabla f(x) + \nabla^2 f(x)v + A^T w = 0, \quad A(x+v) = b$$

Newton decrement

$$\lambda(x) = \left(\Delta x_{\text{nt}}^T \nabla^2 f(x) \Delta x_{\text{nt}} \right)^{1/2} = \left(-\nabla f(x)^T \Delta x_{\text{nt}} \right)^{1/2}$$

properties

- gives an estimate of $f(x) - p^*$ using quadratic approximation $\hat{f}$:

$$f(x) - \inf_{Ay=b} \hat{f}(y) = \frac{1}{2} \lambda(x)^2$$

- directional derivative in Newton direction:

$$\left. \frac{d}{dt} f(x + t \Delta x_{\text{nt}}) \right|_{t=0} = -\lambda(x)^2$$

- in general, $\lambda(x) \neq \left(\nabla f(x)^T \nabla^2 f(x)^{-1} \nabla f(x) \right)^{1/2}$

Newton's method with equality constraints

given starting point $x \in \text{dom } f$ with $Ax = b$, tolerance $\epsilon > 0$.

repeat

1. Compute the Newton step and decrement $\Delta x_{\text{nt}}, \lambda(x)$.
 2. *Stopping criterion.* **quit** if $\lambda^2/2 \leq \epsilon$.
 3. *Line search.* Choose step size t by backtracking line search.
 4. *Update.* $x := x + t\Delta x_{\text{nt}}$.
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- a feasible descent method: $x^{(k)}$ feasible and $f(x^{(k+1)}) < f(x^{(k)})$
- affine invariant

Newton's method and elimination

Newton's method for reduced problem

$$\text{minimize } \tilde{f}(z) = f(Fz + \hat{x})$$

- variables $z \in \mathbf{R}^{n-p}$
- $\hat{x}$ satisfies $A\hat{x} = b$; **rank** $F = n - p$ and $AF = 0$
- Newton's method for $\tilde{f}$, started at $z^{(0)}$, generates iterates $z^{(k)}$

Newton's method with equality constraints

when started at $x^{(0)} = Fz^{(0)} + \hat{x}$, iterates are

$$x^{(k+1)} = Fz^{(k)} + \hat{x}$$

hence, don't need separate convergence analysis

Newton step at infeasible points

2nd interpretation of page 11–6 extends to infeasible x (*i.e.*, $Ax \neq b$)

linearizing optimality conditions at infeasible x (with $x \in \text{dom } f$) gives

$$\begin{bmatrix} \nabla^2 f(x) & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} \Delta x_{\text{nt}} \\ w \end{bmatrix} = - \begin{bmatrix} \nabla f(x) \\ Ax - b \end{bmatrix} \quad (1)$$

primal-dual interpretation

- write optimality condition as $r(y) = 0$, where

$$y = (x, \nu), \quad r(y) = (\nabla f(x) + A^T \nu, Ax - b)$$

- linearizing $r(y) = 0$ gives $r(y + \Delta y) \approx r(y) + Dr(y)\Delta y = 0$:

$$\begin{bmatrix} \nabla^2 f(x) & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} \Delta x_{\text{nt}} \\ \Delta \nu_{\text{nt}} \end{bmatrix} = - \begin{bmatrix} \nabla f(x) + A^T \nu \\ Ax - b \end{bmatrix}$$

same as (1) with $w = \nu + \Delta \nu_{\text{nt}}$

Infeasible start Newton method

given starting point $x \in \text{dom } f$, ν , tolerance $\epsilon > 0$, $\alpha \in (0, 1/2)$, $\beta \in (0, 1)$.

repeat

1. Compute primal and dual Newton steps Δx_{nt} , $\Delta \nu_{\text{nt}}$.

2. *Backtracking line search* on $\|r\|_2$.

$t := 1$.

while $\|r(x + t\Delta x_{\text{nt}}, \nu + t\Delta \nu_{\text{nt}})\|_2 > (1 - \alpha t)\|r(x, \nu)\|_2$, $t := \beta t$.

3. *Update*. $x := x + t\Delta x_{\text{nt}}$, $\nu := \nu + t\Delta \nu_{\text{nt}}$.

until $Ax = b$ and $\|r(x, \nu)\|_2 \leq \epsilon$.

- not a descent method: $f(x^{(k+1)}) > f(x^{(k)})$ is possible
- directional derivative of $\|r(y)\|_2$ in direction $\Delta y = (\Delta x_{\text{nt}}, \Delta \nu_{\text{nt}})$ is

$$\left. \frac{d}{dt} \|r(y + t\Delta y)\|_2 \right|_{t=0} = -\|r(y)\|_2$$

Solving KKT systems

$$\begin{bmatrix} H & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} v \\ w \end{bmatrix} = - \begin{bmatrix} g \\ h \end{bmatrix}$$

solution methods

- LDL^T factorization
- elimination (if H nonsingular)

$$AH^{-1}A^Tw = h - AH^{-1}g, \quad Hv = -(g + A^Tw)$$

- elimination with singular H : write as

$$\begin{bmatrix} H + A^TQA & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} v \\ w \end{bmatrix} = - \begin{bmatrix} g + A^TQh \\ h \end{bmatrix}$$

with $Q \succeq 0$ for which $H + A^TQA \succ 0$, and apply elimination

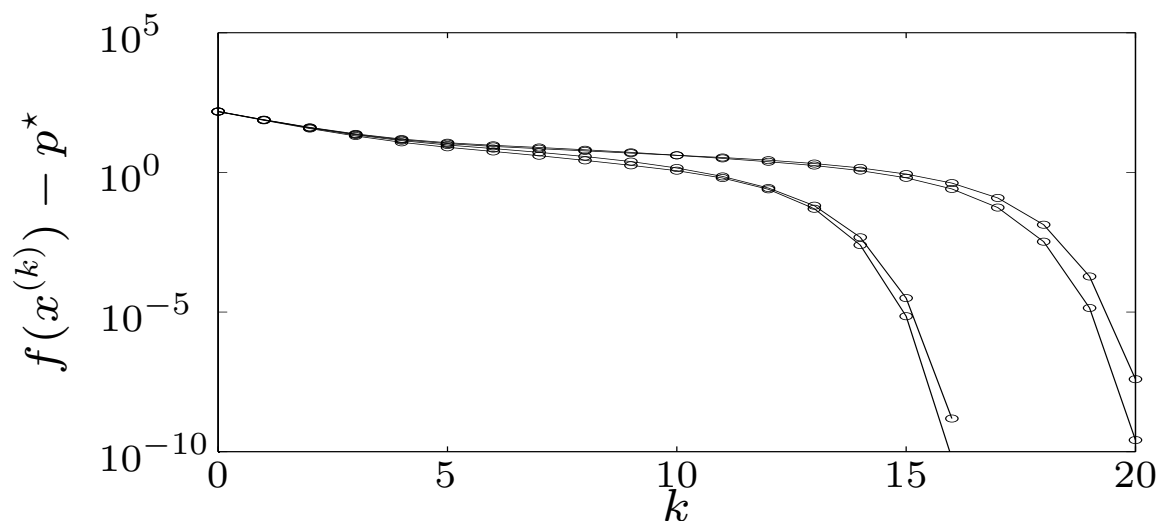
Equality constrained analytic centering

primal problem: minimize $-\sum_{i=1}^n \log x_i$ subject to $Ax = b$

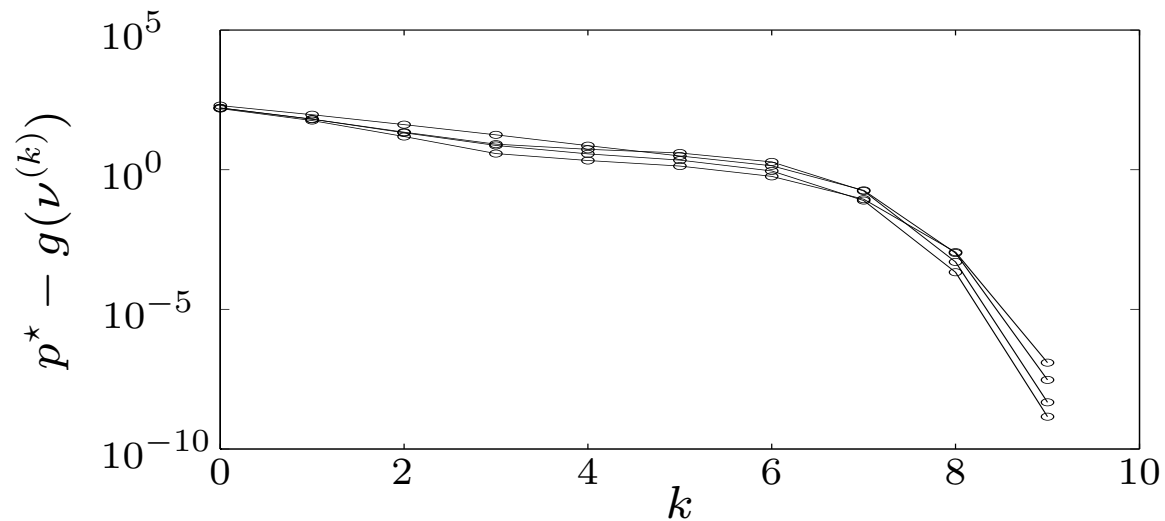
dual problem: maximize $-b^T \nu + \sum_{i=1}^n \log(A^T \nu)_i + n$

three methods for an example with $A \in \mathbf{R}^{100 \times 500}$, different starting points

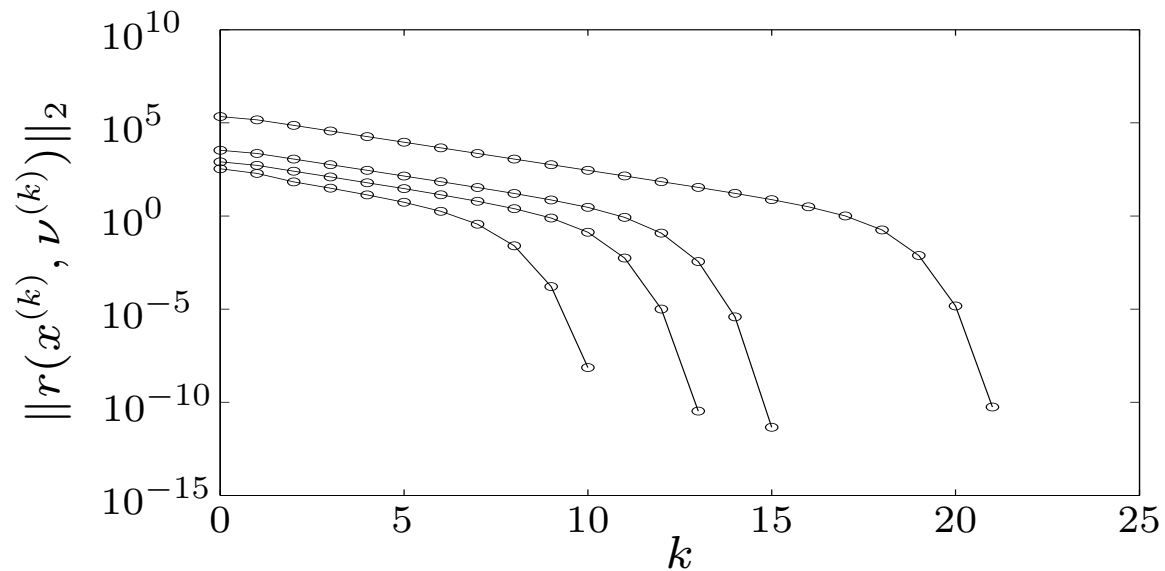
1. Newton method with equality constraints (requires $x^{(0)} \succ 0$, $Ax^{(0)} = b$)



2. Newton method applied to dual problem (requires $A^T \nu^{(0)} \succ 0$)



3. infeasible start Newton method (requires $x^{(0)} \succ 0$)



complexity per iteration of three methods is identical

1. use block elimination to solve KKT system

$$\begin{bmatrix} \mathbf{diag}(x)^{-2} & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} \Delta x \\ w \end{bmatrix} = \begin{bmatrix} \mathbf{diag}(x)^{-1} \mathbf{1} \\ 0 \end{bmatrix}$$

reduces to solving $A \mathbf{diag}(x)^2 A^T w = b$

2. solve Newton system $A \mathbf{diag}(A^T \nu)^{-2} A^T \Delta \nu = -b + A \mathbf{diag}(A^T \nu)^{-1} \mathbf{1}$

3. use block elimination to solve KKT system

$$\begin{bmatrix} \mathbf{diag}(x)^{-2} & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta \nu \end{bmatrix} = \begin{bmatrix} \mathbf{diag}(x)^{-1} \mathbf{1} - A^T \nu \\ b - Ax \end{bmatrix}$$

reduces to solving $A \mathbf{diag}(x)^2 A^T w = 2Ax - b$

conclusion: in each case, solve $ADA^T w = h$ with D positive diagonal